Isaac Reid

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Research interests

I'm a research scientist at Google DeepMind, working on pretraining for Gemini Robotics. I'm broadly interested in making transformers better, especially when learning from multimodal data and with limited access to compute. My previous research focussed on efficient alternatives to softmax attention and novel RoPE-like position encodings, as well as training data attribution in the presence of stochasticity.

Education

Machine Learning, University of Cambridge, UK *Oct 2022 - Feb 2026*
PhD in Engineering
Supervisors: Dr Adrian Weller and Prof. Rich Turner
Advisor: Prof. Carl Rasmussen
Subject: Scalable and data-efficient machine learning. Close collaboration with Prof. Krzysztof Choromanski.

Physics, University of Oxford, UK *Oct 2017 - Jun 2021*
Master of Physics, MPhys
Grade: First class, 92%, top of Oxford cohort (out of ~ 150)
Research project: Quantum entanglement barriers in dual-unitary circuits
Supervisor: Dr Bruno Bertini

Experience

Research Scientist, Google DeepMind (London, UK) *April 2026 - present*
Gemini Robotics

Student Researcher, Google DeepMind (NYC, US and London, UK) *May 2025 - Dec 2025*
Contributing Gemini Robotics 1.5, a new family of robotics foundation models, focusing on advanced embodied reasoning and cross-embodiment transfer

Student Researcher, Google (London, UK) *May 2024 - Nov 2024*
Developed and benchmarked new mechanisms for subquadratic attention for multiple data modalities, applicable to large language model scaling

Research Associate, IQ Capital (London, UK) *Nov 2024 - April 2025*
1 day/week advising and investing in ML startups and supporting portfolio companies

Systems Engineer, Opsydia (Oxford, UK) *Sep 2021 - Sep 2022*
R&D engineer at deep-tech startup specialising in laser technology and adaptive optics, spun out of Oxford University Engineering Department

Research Intern, Max Planck Institute (remote, UK) *Summer 2020*
Computational study of Bose-Einstein condensation in active matter, applying theoretical results from many-body quantum physics to classical clustering phenomena observed in Kob-Andersen particle dynamics
Supervisors: Dr Benoit Mahault and Prof. Ramin Golestanian

Research Intern, Rudolf Peierls Centre for Theoretical Physics (Oxford, UK) *Summer 2019*
Study of relationship between spectral properties of Hessian of loss function and Bayesian prior upon deep neural network initialisation, estimated using random sampling of weights and Gaussian processes
Supervisor: Prof. Ard Louis

Select publications and preprints

Preferred papers

Gemini Robotics 1.5: Pushing the Frontier of Generalist Robots with Advanced Embodied Reasoning, Thinking, and Motion Transfer
GDM blog
GDM robotics team (very many authors)
Synopsis: A new family of robotics **foundation models**, with improved **reasoning** and cross-embodiment transfer
Gemini Robotics 1.5 Tech Report

Rotary Position Encodings for Graphs
ICML 2026, accepted as spotlight paper
Isaac Reid*, Arijit Sehanobish*, Cederik Höfs*, Bruno Mlodozeniec, Leonhard Vulpius, Federico Barbero, Adrian Weller, Krzysztof Choromanski, Richard E. Turner, Petar Veličković
Synopsis: You can extend rotary position encodings (**RoPE**) to graphs, and it automatically inherits a bunch of nice properties
https://arxiv.org/abs/2509.22259

Distributional Training Data Attribution
NeurIPS 2025, accepted as spotlight paper
Bruno Mlodozeniec*, **Isaac Reid***, Sam Power, David Krueger, Murat Erdogdu, Richard Turner, Roger Grosse
Synopsis: 'Influence' of samples can be understood by how drastically the **distribution over trained models** changes if they are removed. Influence functions emerge organically from this new mathematical formulation.
https://arxiv.org/abs/2506.12965

Learning the RoPEs: Better 2D and 3D Position Encodings with STRING
ICML 2025, accepted as spotlight paper
Connor Schenck*, **Isaac Reid***, Mithun George Jacob*, Alex Bewley*, Joshua Ainslie*, David Rendleman*, Deepali Jain, Mohit Sharma, Avinava Dubey, Ayzaan Wahid, Sumeet Singh, René Wagner, Tianli Ding, Chuyuan Fu, Arunkumar Byravan, Jake Varley, Alexey Gritsenko, Matthias Minderer, Dmitry Kalashnikov, Jonathan Tompson, Vikas Sindhwani, Krzysztof Choromanski*
Synopsis: Separable Translationally Invariant Position Encodings = STRING. A more general extension of rotary position encodings (**RoPE**) using some very lightweight group theory, applicable to images and videos
https://arxiv.org/abs/2506.12965

Simplex Random Features
ICML 2023, accepted with oral presentation
Isaac Reid, Krzysztof Choromanski, Valerii Likhosherstov, Adrian Weller
Synopsis: Derivation of an optimal random feature mechanism for unbiased approximation of the Gaussian kernel, motivated by a host of new analytical results and tested with **subquadratic transformer** experiments
https://arXiv.org/abs/2301.13856

Entanglement Barriers in Dual-Unitary Circuits
Phys. Rev. B 104, 014301 – Published 1 July 2021
Isaac Reid, Bruno Bertini
Synopsis: Exact characterisation of the dynamics of quantum entanglement arising after a quantum quench in a many-body, locally interacting system, including both the integrable and completely chaotic regimes
https://arxiv.org/abs/2103.12794

Other papers

Gaussian Random Features for Scalable Gaussian Processes
ICLR 2026
Matthew Zhang, Jihao Andreas Lin, Krzysztof Choromanski, Adrian Weller, Richard Turner, **Isaac Reid**
Synopsis: Graph random features permit very efficient Bayesian optimisation on graphs, when you implement the sparse linear algebra properly and use iterative linear solvers instead of matrix inverses
https://arxiv.org/abs/2509.03691

Linear Transformer Topological Masking with Graph Random Features
ICLR 2025
Isaac Reid, Kumar Avinava Dubey, Deepali Jain, Will Whitney, Amr Ahmed, Joshua Ainslie, Alex Bewley, Mithun Ja-

cob, Aranyak Mehta, David Rendleman, Connor Schenck, Richard E. Turner, René Wagner, Adrian Weller, Krzysztof Choromanski
Synopsis: You can use graph random features to efficiently incorporate information about the underlying graph structure into a linear attention transformer, without instantiating attention
https://arxiv.org/abs/2410.03462

Optimal Time Complexity Algorithms for Computing General Random Walk Graph Kernels on Sparse Graphs
AISTATS 2025
Krzysztof Choromanski*, **Isaac Reid***, Arijit Sehanobish, Avinava Dubey
Synopsis: By simulating correlated random walks on an ensemble of graphs, you can estimate the graph kernel between any given pair in linear time
https://arxiv.org/abs/2410.10368

Variance-Reducing Couplings for Random Features
ICLR 2025
Isaac Reid, Stratis Markou, Krzysztof Choromanski, Richard E. Turner, Adrian Weller
Synopsis: Variance reduction in Monte Carlo is really a multi-marginal optimal transport problem, and treating it as such gives us tools to sample more efficiently in Euclidean and discrete space
https://arxiv.org/abs/2405.16541

General Graph Random Features
ICLR 2024
Isaac Reid*, Krzysztof Choromanski*, Eli Berger*, Adrian Weller
Synopsis: You give me an arbitrary function of a weighted adjacency matrix, I give you a random feature mechanism to approximate it efficiently
https://arxiv.org/abs/2310.04859

Repelling Random Walks
ICLR 2024
Isaac Reid, Eli Berger, Krzysztof Choromanski, Adrian Weller
Synopsis: Now we correlate walker directions in an algorithm of broader interest
https://arxiv.org/abs/2310.04854

Quasi-Monte Carlo Graph Random Features
NeurIPS 2023, accepted as spotlight paper
Isaac Reid, Krzysztof Choromanski, Adrian Weller
Synopsis: A QMC scheme that induces correlations between the lengths random walks on a graph
https://arxiv.org/abs/2305.12470

Teaching

Cambridge MLMI *Lent 2025*
Synopsis: Guest lecturer, providing an introduction to transformers.

Engineering 2P7 *Michaelmas 2023 - Easter 2025*
Synopsis: Supervisions in mathematics for engineers (vector calculus, linear algebra and probability)

Pembroke International Summer Programme *Jun-Aug 2023*
Synopsis: Research project on density ratio estimation in machine learning

Select talks

Efficient Transformers – Machine Learning Group, Cambridge *Nov 2024*
Synopsis: Seminar on mathematical tricks for subquadratic attention

Quasi-Monte Carlo Graph Random Features – NeurIPS@Cambridge, Cambridge *Dec 2023*
Synopsis: Invited talk to accompany NeurIPS spotlight paper

Simplex Random Features – ICML 2023, Honolulu *July 2023*
Synopsis: Oral presentation to accompany paper

Simplex Random Features – Microsoft Research, Cambridge *Jun 2023*
Synopsis: Research talk on ICML paper

Random Features for Kernel Approximation – Machine Learning Group, Cambridge *Mar 2023*
Synopsis: Seminar on random feature methods and recent QMC schemes to improve their convergence

Scholarships and awards

Google PhD Fellowship in Algorithms and Theory *2024-2026*
Funding from Google for remainder of PhD

G-Research Grant *July 2023*
Financial award to help fund attendance of ICML conference

Trinity College External Studentship *2022-2025*
Full scholarship for a PhD in Machine Learning

Encaenia *Jun 2022*
One of six undergraduate students invited to attend Oxford's historic Encaenia ceremony

Gibbs Prize *2020-2021*
For submitting the highest scoring MPhys research project (87%)

Scott Prize *2017-2021*
For best overall performance in the MPhys (92%)

Scott Prize *2019-2020*
For best performance in the third year (92%)

Winton Capital Prize *2018-2019*
For best performance in the second year (93%)

Hertford College Academic Scholarship *2018-2021*
For performance in first year (88%)